

Prodip Sengupta

AI-Powered Full-Stack Developer · MERN · LLM Agents · RAG Pipelines · Generative AI

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PROFILE

Full-stack developer specialising in MERN and Generative AI, with production experience building multi-agent systems, RAG pipelines, and tool-calling LLM applications using Claude, GPT-4o, and Gemini. Built and deployed end-to-end AI products integrating LLM agents, PostgreSQL, MongoDB, and cloud infrastructure. Open to full-stack and AI engineer roles.

EXPERIENCE

Frontend Developer & Project Manager

Nov 2025 – Apr 2026

Tara Effects Photography · Freelance · tara-effects.vercel.app

- Delivered a production Next.js website for a wedding photography client covering photography, videography, and digital invitation services — built with Framer Motion scroll-triggered transitions and dark/light theme toggle.
- Achieved Lighthouse performance score of 95+ on mobile; integrated Cloudinary for optimised image delivery (avg load < 1 s) and WhatsApp Business API for zero-friction client booking.
- Owned full end-to-end delivery as Project Manager: requirements gathering, design sign-off, and Vercel deployment.

PROJECTS

FinDoc AI — RAG-Powered SEC Filing Research Assistant

Jun 2026

React · Node.js · Express · LangGraph JS · Supabase · pgvector · OpenAI

[Live Demo](#) | [GitHub](#)

- Built a full-stack RAG assistant enabling investment analysts to query thousands of pages of SEC filings (10-Ks, 10-Qs, earnings transcripts) in natural language, with every answer grounded in cited source passages.
- Engineered a hybrid retrieval pipeline combining **pgvector semantic search** and **Postgres full-text search**, fused via Reciprocal Rank Fusion — reducing analyst document-intake time **from hours to seconds**.
- Orchestrated a **LangGraph ReAct agent** that reasons over retrieved chunks, calls retrieval tools iteratively, and enforces structured JSON output with numbered citation markers tied to exact document UUIDs.

GAIA Benchmark Agent — Stateful Multi-Step LLM Reasoner

May 2026

LangGraph JS · Node.js · Gemini 2.5 Flash · GPT-4o (Azure) · Qwen-2.5 · Docker

[Hugging Face Space](#) | [GitHub](#)

- Built a stateful cyclic multi-step reasoning agent using LangGraph JS to autonomously solve general assistant tasks on the GAIA benchmark.
- Engineered a tool orchestration layer integrating sandboxed Python execution, web grounding, Wayback Machine history, and multimodal image analysis — dynamically parsing text, table, audio, and visual attachments.
- Implemented a resilient model fallback pool (Gemini → GPT-4o → Qwen) to recover from rate limits; achieved **35% accuracy on Level-1 validation tasks** with a Dockerized HF Spaces deployment.

LeafCart — AI-Powered Grocery Delivery Platform

Mar 2026

MERN · Gemini Function Calling · Stripe · JWT

[Live Demo](#) | [GitHub](#)

- Built a full-stack MERN e-commerce platform serving **50+ products** across categories, with dynamic filtering, cart management, address handling, and dual seller/admin dashboards for real-time inventory and order tracking.
- Engineered an AI cart agent via **Gemini function calling** that interprets natural-language requests (e.g. “I want to cook Palak Paneer”), autonomously queries MongoDB, and maintains isolated per-user session memory.
- Secured with JWT auth and Stripe payments; deployed on Vercel with sub-second API responses (**avg ~600 ms**) and a horizontally scalable MongoDB backend.

SKILLS

Languages: JavaScript (ES2024), Python, Java, HTML5, CSS3

Frameworks: React.js, Node.js, Express.js, Tailwind CSS, Redux

Generative AI: OpenAI GPT-4o, Gemini, Claude / Anthropic, LangChain, LangGraph, Prompt Engineering, RAG Pipelines, AI Agents, Function Calling

Databases: MongoDB, PostgreSQL, Supabase, pgvector, Pinecone (Vector DB)

Tools: Git, GitHub, Postman, Vercel, Netlify, Cloudinary, Stripe API, Docker, Framer Motion

EDUCATION

Master of Computer Applications Maulana Abul Kalam Azad University of Technology · CGPA 7.80 / 10	Aug 2025
Bachelor of Computer Applications University of North Bengal · CGPA 9.07 / 10	Jul 2023

CERTIFICATIONS

Claude Code in Action — Anthropic (Verified)	Mar 2026
Claude 101 — Anthropic (Verified)	Mar 2026
AI Agents Course (Certificate of Excellence) — Hugging Face (Verified)	May 2026
Introduction to Model Context Protocol — Anthropic (Verified)	Jun 2026